Numerical methods

Lecture 4: LU decomposition, vector norms

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Recall the LU decomposition

Definition: LU decomposition

The product $L \cdot U$ is called the LU decomposition of the matrix A, if

$$A = LU, \quad L \in \mathcal{L}_1, \;\; U \in \mathcal{U}.$$

Why is the LU decomposition useful?

Assume that

- the SLE $Ax = b$ LER has a solution,
- we have the $A = LU$ decomposition.

Then instead of $Ax = L \cdot \underbrace{U \cdot x}_{y} = b$ solve the systems

$(\frac{2}{3}n^3 + \mathcal{O}(n^2))$

1. $Ly = b$ (lower triangular) and $(n^2 + \mathcal{O}(n))$
2. $Ux = y$ (upper triangular). $(n^2 + \mathcal{O}(n))$

For comparison, for a matrix-vector product:
$n \cdot (2n - 1) = 2n^2 + \mathcal{O}(n)$.

Of course the LU decomposition must be formed. $(\frac{2}{3}n^3 + \mathcal{O}(n^2))$
Advantageous if A is the same for many SLE.

Direct calculation of the LU decomposition

- L and U not known: unknown in the matrices.
- Their product is known: $LU = A$.
- Writing the elements of A, we get equations for the elements of L and U.
- Choosing a *good order* for the equations, we always find a new unknown.
- Discussing GE, we saw that row 1 of U is the same as row 1 of A
  (GE does not change the first row).
- The first column of L is the first column of A divided by a_{11}.

Good orders

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 3 & 3 \\ 4 & 4 & 5 & 5 \\ 6 & 6 & 6 & 7 \end{pmatrix}$$

row-wise

$$\begin{pmatrix} 1 & 3 & 5 & 7 \\ 2 & 3 & 5 & 7 \\ 2 & 4 & 5 & 7 \\ 2 & 4 & 6 & 7 \end{pmatrix}$$

column-wise

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 3 & 3 \\ 2 & 4 & 5 & 5 \\ 2 & 4 & 6 & 7 \end{pmatrix}$$

parquet-like

Example: LU decomposition directly

a. Find the LU decomposition of our sample matrix directly, based on matrix multiplication.

b. Consider a new example too. (Careful, $\det(B_2) = 0$.)

$$A = \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & -2 & 3 \\ -4 & 4 & -2 \\ 6 & -5 & 4 \end{bmatrix}$$

Row-wise: Row 1 of U is known. For Row 2:

$$\begin{bmatrix} 2 & 0 & 3 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \qquad \begin{aligned} l_{21} \cdot 2 &= -4 \\ l_{21} \cdot 0 + 1 \cdot u_{22} &= 5 \\ l_{21} \cdot 3 + 1 \cdot u_{23} &= -2 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad \begin{bmatrix} 2 & 0 & 3 \\ -4 & 5 & -2 \\ 6 & -5 & 4 \end{bmatrix} \qquad \begin{aligned} l_{21} &= -2 \\ u_{22} &= 5 \\ u_{23} &= -2 - (-2) \cdot 3 = 4 \end{aligned}$$

For Row 3:

$$\begin{aligned} l_{31} \cdot 2 &= 6 \\ l_{31} \cdot 0 + l_{32} \cdot u_{22} &= -5 \\ l_{31} \cdot 3 + l_{32} \cdot u_{23} + 1 \cdot u_{33} &= 4 \end{aligned} \qquad \begin{aligned} l_{31} &= 3 \\ l_{32} &= \frac{-5}{5} = -1 \\ u_{33} &= 4 - 3 \cdot 3 - (-1) \cdot 4 = -1 \end{aligned}$$

Row-wise: Row 1 of U is known. For Row 2:

$$\begin{bmatrix} 2 & -2 & 3 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \qquad \begin{aligned} l_{21} \cdot 2 &= -4 \\ l_{21} \cdot (-2) + 1 \cdot u_{22} &= 4 \\ l_{21} \cdot 3 + 1 \cdot u_{23} &= -2 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad \begin{bmatrix} 2 & -2 & 3 \\ -4 & 4 & -2 \\ 6 & -5 & 4 \end{bmatrix} \quad \begin{aligned} l_{21} &= -2 \\ u_{22} &= 4 - (-2) \cdot (-2) = 0 \\ u_{23} &= -2 - (-2) \cdot 3 = 4 \end{aligned}$$

For Row 3:

$$l_{31} \cdot 2 = 6 \qquad l_{31} = 3$$

$$l_{31} \cdot (-2) + l_{32} \cdot u_{22} = -5 \qquad \rightsquigarrow \text{inconsistent}$$

Since $D_2 = \det(B_2) = 0$, we get $u_{22} = 0$. Can't find LU decomposition. With GE: $a_{22}^{(1)} = 0$, switch rows.

Direct calculation of the LU decomposition

Theorem: direct calculation of the LU decomposition

The elements of the matrices L and U can be calculated as:

$$i \le j \text{ (upper)} \qquad u_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} \cdot u_{kj},$$

$$i > j \text{ (lower)} \qquad l_{ij} = \frac{1}{u_{jj}} \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} \cdot u_{kj} \right).$$

Calculating in a good order, the right-hand-side is always known.

Direct calculation of the LU decomposition

Proof. Write the element $A_{i,j}$ of the matrix $A \in \mathbb{R}^{n \times n}$ from the matrix product, assuming $A = L \cdot U$. Use the face that we have triangular matrices. Separate one term.

If $i \leq j$, i.e. for an element above (or in) the diagonal, then $k > i \Rightarrow l_{ik} = 0$, and $l_{ii} = 1$, thus

$$a_{ij} = \sum_{k=1}^{n} l_{ik} \cdot u_{kj} = \sum_{k=1}^{i} l_{ik} \cdot u_{kj} = u_{ij} + \sum_{k=1}^{i-1} l_{ik} \cdot u_{kj}.$$

Express u_{ij}

$$u_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} \cdot u_{kj}.$$

Direct calculation of the LU decomposition

If $i > j$, i.e. for an element below the diagonal, then
$k > j \Rightarrow u_{kj} = 0$, thus

$$a_{ij} = \sum_{k=1}^{n} l_{ik} \cdot u_{kj} = \sum_{k=1}^{j} l_{ik} \cdot u_{kj} = l_{ij} \cdot u_{jj} + \sum_{k=1}^{j-1} l_{ik} \cdot u_{kj}.$$

If $u_{jj} \neq 0$ (we met this before...), then express l_{ij}

$$l_{ij} = \frac{1}{u_{jj}} \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} \cdot u_{kj} \right).$$

Observe that using 'good orders' for traversing the (i, j) indices, we found that the entire right-hand-side is known.

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Operation count of the LU decomposition

Theorem: Operation count of the LU decomposition

$$\frac{2}{3}n^3 + \mathcal{O}(n^2)$$

Proof. Trivially the same as of GE, since it can be used to find the LU decomposition. □

Remark. You may practice finding the operation count according to the formulas of the 'direct' calculation.

Theorem: operation count to solve $Ux = y$

$$n^2 + \mathcal{O}(n)$$

Proof. See back substitution (after GE).

Theorem: operation count to solve $Ly = b$

$$n^2 + \mathcal{O}(n)$$

Proof. Instead of from x_n to x_1, we can solve from x_1 to x_n. Try to write the formulas to practice...

Contents

Definition: symmetric matrices

The matrix A is *symmetric*, if $A = A^\top$.

Definition: positive definite matrices

The symmetric matrix $A \in \mathbb{R}^{n\times n}$ is *positive definite*, if

1. $\langle Ax, x\rangle = x^\top Ax > 0$ for all $0 \neq x \in \mathbb{R}^n$; or
2. for all principal minors $D_k = \det(A_k) > 0$; or
3. all eigenvalues of A are positive.

Proposition: equivalent description of positive definite matrices

The below conditions **1. 2. 3.** are equivalent.

Proof. Skipped. $\square$

Definition

The matrix $A \in \mathbb{R}^{n \times n}$ is *strictly diagonally dominant by its rows*, if $|a_{ii}| > \sum_{j=1, j \neq i} |a_{ij}| \quad (i = 1, \ldots, n)$.

Definition

The matrix $A \in \mathbb{R}^{n \times n}$ is *strictly diagonally dominant by its columns*, if $|a_{ii}| > \sum_{j=1, j \neq i} |a_{ji}| \quad (i = 1, \ldots, n)$.

Example

$$\begin{bmatrix} 4 & 1 & -2 \\ -2 & 5 & 1 \\ 0 & -3 & 4 \end{bmatrix}$$

Definition

The *half-bandwidth* of the matrix A is $s \in \mathbb{N}$, if

$$\forall\ i,j : |i-j| > s : \ \ a_{ij} = 0 \text{ and}$$
$$\exists\ k,l : |k-l| = s : a_{kl} \neq 0.$$

Example

The below matrix is symmetric, positive definite and its half-bandwidth is 1.

$$\begin{bmatrix} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

Definition

The *profiles* of the matrix A by rows and by columns are the n tuples $(k_1, \ldots, k_n)$ and $(l_1, \ldots, l_n)$ respectively, such that

$$\forall \ j = 1, \ldots, k_i : \ \ a_{ij} = 0 \ \text{és} \ a_{i,k_i+1} \neq 0,$$
$$\forall \ i = 1, \ldots, l_j : \ \ a_{ij} = 0 \ \text{és} \ a_{l_j+1,j} \neq 0.$$

Example

Profiles: $(0, 0, 2, 1)$ by rows, and $(0, 1, 1, 2)$ by columns.

$$\begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & 4 & 1 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 1 & 2 & 4 \end{bmatrix}$$

Partition the SLE $Ax = b$ after the kth row ($k < n$, $k \in \mathbb{N}$) assuming that $A_{11} \in \mathbb{R}^{k \times k}$ is invertible.

$$\left[\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right] \cdot \left[\begin{array}{c} x_1 \\ \hline x_2 \end{array}\right] = \left[\begin{array}{c} b_1 \\ \hline b_2 \end{array}\right]$$

The system in the partitioned form:

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 &= b_1 \\ A_{21}x_1 + A_{22}x_2 &= b_2 \end{aligned}$$

Perform a step of *block* GE:
Equation 2 $-$ $(A_{21} \cdot A_{11}^{-1}) \cdot$ Equation 1

$$\underbrace{\left(A_{21} - A_{21}A_{11}^{-1}A_{11}\right)}_{0} x_1 + \left(A_{22} - A_{21}A_{11}^{-1}A_{12}\right) x_2 = b_2 - A_{21}A_{11}^{-1}b_1$$

Schur complement

Equation 2 after the block GE step:

$$\left(A_{22} - A_{21}A_{11}^{-1}A_{12}\right) x_2 = b_2 - A_{21}A_{11}^{-1}b_1.$$

Partitioning in matrix form:

$$\left[\begin{array}{r|c} A_{11} & A_{12} \\ \hline 0 & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{array}\right] \cdot \left[\begin{array}{c} x_1 \\ \hline x_2 \end{array}\right] = \left[\begin{array}{c} b_1 \\ \hline b_2 - A_{21}A_{11}^{-1}b_1 \end{array}\right]$$

- GE is continued on the $(n-k) \times (n-k)$ lower right part.
- Setting $k = 1$ we have $A_{11} = (a_{11})$. Assuming that $a_{11} \neq 0$, then we get back the (regular, no-block) GE step.

Definition: Schur complement

Assume that $A_{11} \in \mathbb{R}^{k \times k}$ is invertible. The *Schur complement* of the matrix A with respect to the block A_{11} is the matrix

$$[A|A_{11}] := A_{22} - A_{21}A_{11}^{-1}A_{12}$$

of size $(n-k) \times (n-k)$.

Remarks:

- Schur complement: the matrix to continue the GE with.
- Useful to state invariant properties, preserved/inherited by GE

Theorem: conservation laws for GE

The following properties of the matrix A are preserved during Gaussian elimination, are inherited to the Schur complement:

1. $\det(A) \neq 0 \quad \Rightarrow \quad \det([A|A_{11}]) \neq 0$
2. A is symmetric $\Rightarrow [A|A_{11}]$ is symmetric
3. A is positive definite $\Rightarrow [A|A_{11}]$ is positive definite
4. A str. diag. dom. $\Rightarrow [A|A_{11}]$ str. diag. dom.
5. the half-bandwith of $[A|A_{11}] \leq$ the half-bandwidth if A

Remark. Consider these with respect to the LU decomposition. What about the profile?

Proof. (1) Determinant:
Since the GE preserves the determinant, we have
$\det(A) = \det(A^{(1)}) \neq 0$.

$$A^{(1)} = \left[\begin{array}{r|c} A_{11} & A_{12} \\ \hline 0 & [A|A_{11}] \end{array}\right]$$

$$0 \neq \det(A^{(1)}) = \underbrace{\det(A_{11})}_{\neq 0} \cdot \det([A|A_{11}]) \quad \Leftrightarrow \quad \det([A|A_{11}]) \neq 0$$

$\square$

(2) Symmetry:
If A is symmetric, then so is A_{11} and A_{22}, furthermore $A_{21}^\top = A_{12}$.

$$\begin{aligned} [A|A_{11}]^\top &= (A_{22} - A_{21}A_{11}^{-1}A_{12})^\top = A_{22}^\top - A_{12}^\top (A_{11}^{-1})^\top A_{21}^\top = \\ &= A_{22}^\top - A_{12}^\top (A_{11}^\top)^{-1} A_{21}^\top = A_{22} - A_{21}A_{11}^{-1}A_{12} = [A|A_{11}] \end{aligned}$$

$\square$

Proofs for (3), (4), (5) skipped.

Also known as: Tridiagonal matrix algorithm, Thomas algorithm. Tridiagonal systems are very common, a significantly more efficient algorithm exists.

- Storage: instead of n^2, we have $3n-2$ elements in 3 diagonals.
- Operation count: instead of $\frac{2}{3}n^3 + \mathcal{O}(n^2)$, only $8n + \mathcal{O}(1)$.

During GE, the Schur complement stays tridiagonal.
After GE, U has two diagonals, thus in the back substitution we have equations of the form:

$$a_{ii}^{(i-1)} x_i + a_{ii+1}^{(i-1)} x_{i+1} = a_{in+1}^{(i-1)}.$$

Expressing x_i we have $x_i = f_i x_{i+1} + g_i$ $(i = 1, \ldots, n)$ with some f_i, g_i values.

Notation: $A = \text{tridiag}\,(\beta_{i-1}, \alpha_i, \gamma_i)$,

$$A = \begin{bmatrix} \alpha_1 & \gamma_1 & 0 & \cdots & 0 \\ \beta_1 & \alpha_2 & \gamma_2 & & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \beta_{n-2} & \alpha_{n-1} & \gamma_{n-1} \\ 0 & & 0 & \beta_{n-1} & \alpha_n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix}.$$

Equation 1:

$$\alpha_1 x_1 + \gamma_1 x_2 = b_1 \quad \rightarrow \quad \alpha_1 x_1 = -\gamma_1 x_2 + b_1 \quad \rightarrow \quad x_1 = -\frac{\gamma_1}{\alpha_1} x_2 + \frac{b_1}{\alpha_1}$$

In the form $x_1 = f_1 x_2 + g_1$ we have $f_1 = -\dfrac{\gamma_1}{\alpha_1}$ és $g_1 = \dfrac{b_1}{\alpha_1}$.

Assume that $f_1, \ldots, f_{i-1}$ and $g_1, \ldots, g_{i-1}$ are known for the recursion steps $x_k = f_k x_{k+1} + g_k$ $(k = 1, \ldots, i-1)$.
We are to find $x_i = f_i x_{i+1} + g_i$.
Write equation i and substitute x_{i-1} with the recursive formula:

$$\begin{aligned}
\beta_{i-1} x_{i-1} + \alpha_i x_i + \gamma_i x_{i+1} &= b_i \\
\beta_{i-1}(f_{i-1} x_i + g_{i-1}) + \alpha_i x_i + \gamma_i x_{i+1} &= b_i \\
(\beta_{i-1} f_{i-1} + \alpha_i) x_i + \gamma_i x_{i+1} &= b_i - \beta_{i-1} g_{i-1}
\end{aligned}$$

$$\begin{aligned}
(\alpha_i + \beta_{i-1} f_{i-1}) x_i &= -\gamma_i x_{i+1} + (b_i - \beta_{i-1} g_{i-1}) \\
x_i &= \underbrace{-\frac{\gamma_i}{\alpha_i + \beta_{i-1} f_{i-1}}}_{f_i} x_{i+1} + \underbrace{\frac{b_i - \beta_{i-1} g_{i-1}}{\alpha_i + \beta_{i-1} f_{i-1}}}_{g_i}.
\end{aligned}$$

For the last (nth equation), substituting x_{n-1}:

$$\begin{aligned} \beta_{n-1}x_{n-1} + \alpha_n x_n &= b_n \\ \beta_{n-1}(f_{n-1}x_n + g_{n-1}) + \alpha_n x_n &= b_n \\ (\beta_{n-1}f_{n-1} + \alpha_n)x_n &= b_n - \beta_{n-1}g_{n-1} \end{aligned}$$

$$x_n = \frac{b_n - \beta_{n-1}g_{n-1}}{\alpha_n + \beta_{n-1}f_{n-1}} =: g_n$$

$\square$

Algorithm: progonka method

1st phase: $f_1 := -\frac{\gamma_1}{\alpha_1}, \quad g_1 := \frac{b_1}{\alpha_1}$

$$i = 2, \ldots, n-1: \quad f_i := -\frac{\gamma_i}{\alpha_i + \beta_{i-1} f_{i-1}}$$

$$g_i := \frac{b_i - \beta_{i-1} g_{i-1}}{\alpha_i + \beta_{i-1} f_{i-1}}$$

$$g_n := \frac{b_i - \beta_{n-1} g_{n-1}}{\alpha_n + \beta_{n-1} f_{n-1}}$$

2nd phase: $x_n := g_n$

$i = n-1, n-2, \ldots, 1: \quad x_i = f_i x_{i+1} + g_i$

Remark. 3 operations more, but less formulas, if we find also f_n, and start phase 2 with $x_{n+1} := 0$. Find operation count!

Contents

Recall: 'length' of vectors

Definition: length of a vector

The (usual) length, or 'two-norm' of the vector $x \in \mathbb{R}^n$ is denoted by $\|.\|_2$. It is calculated as:

$$\|x\|_2 := \sqrt{\langle x, x \rangle} = \sqrt{x^\top x} = \left(\sum_{k=1}^{n} x_i^2 \right)^{\frac{1}{2}}.$$

The *(vector) norm* is a generalization of 'length', 'size'.

Definition: vector norm

Fix $n \in \mathbb{N}$. The map $\|.\| : \mathbb{R}^n \to \mathbb{R}$ is called a *vector norm*, if:

1. $\|x\| \geq 0 \quad (\forall x \in \mathbb{R}^n)$,
2. $\|x\| = 0 \iff x = 0$,
3. $\|\lambda \cdot x\| = |\lambda| \cdot \|x\| \quad (\forall \lambda \in \mathbb{R},\ \forall x \in \mathbb{R}^n)$,
4. $\|x + y\| \leq \|x\| + \|y\| \quad (\forall x, y \in \mathbb{R}^n)$.

These are the *axioms* of vector norms.

Proposition: vector norm generated by the scalar product

If the scalar product $\langle .,. \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is available, then the function $f(x) := \sqrt{\langle x, x \rangle}$ is a *norm*, denote by: $\|x\|_2$.

Proof. Skipped.
This is the 'usual length'. $\square$

Proposition: Cauchy–Bunyakovszki–Schwarz inequality (CBS)

$$|\langle x, y\rangle| \leq \|x\|_2 \cdot \|y\|_2 \quad (x, y \in \mathbb{R}^n)$$

Proof. For all $\alpha \in \mathbb{R}$ we have $\|x - \alpha y\|_2^2 \geq 0$.

$$0 \leq \|x - \alpha y\|_2^2 = \langle x - \alpha y, x - \alpha y\rangle =$$
$$= \underbrace{\langle x, x\rangle}_{\|x\|_2^2} - 2\alpha \langle x, y\rangle + \alpha^2 \underbrace{\langle y, y\rangle}_{\|y\|_2^2} \qquad (\forall \alpha \in \mathbb{R}).$$

The discriminant is non-positive: $\langle x, y\rangle^2 - \|x\|_2^2 \cdot \|y\|_2^2 \leq 0$, thus

$$\langle x, y\rangle^2 \leq \|x\|_2^2 \cdot \|y\|_2^2 .$$

$\square$

Proposition: Common vector norms $(1, 2, \infty)$

The below formulas define vector norms on $\mathbb{R}^n$:

- $\|x\|_1 := \sum_{i=1}^{n} |x_i|$ (Manhattan norm),
- $\|x\|_2 := \left(\sum_{i=1}^{n} |x_i|^2\right)^{1/2}$ (Euclidean norm),
- $\|x\|_\infty := \max_{i=1}^{n} |x_i|$ (Chebyshev norm).

Proof. Verify the axioms. Left as an exercise. $\square$

Example

Find the $1, 2, \infty$ norms of the following vectors:

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad y = \begin{bmatrix} 4 \\ -8 \\ 1 \end{bmatrix}.$$

$\|x\|_1 = 3 + 4 = 7$, $\|x\|_2 = \sqrt{3^2 + 4^2} = 5$, $\|x\|_\infty = \max\{3, 4\} = 4$.

$\|y\|_1 = 4 + |-8| + 1 = 13$, $\|y\|_2 = \sqrt{4^2 + (-8)^2 + 1^2} = \sqrt{81} = 9$, $\|y\|_\infty = \max\{4, |-8|, 1\} = 8$.

Proposition: *p*-norms

The below $\mathbb{R}^n \to \mathbb{R}$ functions also define vector norms:

$$\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \qquad (p \in \mathbb{R},\ 1 \leq p < \infty).$$

Proof. Skipped. See also: Minkovszki inequality. $\square$

Remarks:

- $0 \leq p < 1$ will not give a norm,
- $p_1 \leq p_2 \Longrightarrow \|x\|_{p_1} \geq \|x\|_{p_2}$,
- Special cases: $p = 1 \rightsquigarrow \|x\|_1$, $\quad p = 2 \rightsquigarrow \|x\|_2$,
- Furthermore $\lim\limits_{p \to \infty} \|x\|_p = \|x\|_\infty$.

Proposition: inequalities between norms

- $\|x\|_\infty \leq \|x\|_1 \leq n \cdot \|x\|_\infty$,
- $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \cdot \|x\|_\infty$,
- $\|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \cdot \|x\|_2$,
- thus also $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ holds.

Proof. Skipped. $\square$

(The first one is easy, we saw an example for the fourth.)

Definition: equivalent norms

The vector norms $\|.\|_a$ and $\|.\|_b$ are *equivalent*, if $\exists c_1, c_2 \in \mathbb{R}^+$, such that

$$c_1 \cdot \|x\|_b \leq \|x\|_a \leq c_2 \cdot \|x\|_b \qquad (\forall x \in \mathbb{R}^n).$$

Proposition: equivalence of norms in finite dimensions

Any norm defined on $\mathbb{R}^n$ is equivalent to the Euclidean norm. (I.e. all norms are equivalent in a finite dimensional vector space.)

Definition: convergence in norm

The sequence $(x_k) \subset \mathbb{R}^n$ is convergent, if there exists $x^* \in \mathbb{R}^n$, such that

$$\lim_{k \to \infty} \|x_k - x^*\| = 0.$$

x^* is the limit of the sequence.

Remark. Since the vector norms of $\mathbb{R}^n$ are all equivalent, if a sequence is convergent with respect to one norm, then it is with respect to all others too.

Equivalent statements for convergence:

- The sequence $(x_k) \subset \mathbb{R}^n$ is convergent, if $\exists x^* \in \mathbb{R}^n$, such that

$$\forall \, \varepsilon > 0 : \;\; \exists \, N_0 \in \mathbb{N} : \;\; \forall \, k \geq N_0 : \;\; \|x_k - x^*\| < \varepsilon.$$

- The sequence $(x_k) \subset \mathbb{R}^n$ is convergent, if $\exists x^* \in \mathbb{R}^n$, such that

$$\forall \, \varepsilon > 0 : \;\; \exists \, N_0 \in \mathbb{N} : \;\; \forall \, k \geq N_0 : \;\; x_k \in B_\varepsilon(x^*).$$

Examples in Matlab

1. Examples for positive definite matrices.
2. Examples for p norms, unit balls ($p = 1, 2, \infty, \dots$).